

Question Number	Answer	Additional Guidance	Mark
2(a)(i)	An answer that includes: 22.5 breaths per minute (1)	Accept 21 / 24 breaths per minute	(1)

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	An answer including the following: • estimation of tidal volume (1) • estimation of respiratory minute ventilation (1)	Accept a value in the range 0.6 – 0.75 Accept a value in the range 13.2 – 18.0 (Allow ecf) Correct answer with no working gains full marks	(2)

Question Number	Answer	Additional Guidance	Mark
2(a)(iii)	An explanation including two of the following: • more oxygen is needed for increased (aerobic) respiration (1) • because {muscles / muscle cells} need more {ATP / energy} (for contraction) (1) • because more carbon dioxide ({is being produced by respiration / must be removed}) (1)	Accept more oxygen needed for {conversion of lactic acid to pyruvate / oxidation of lactate}	(2)

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	An answer including one abiotic and one biotic factor: Abiotic (1) • (room) temperature • humidity • altitude Biotic (1) • {age / sex} of participant • {health / fitness level} of participant • lung capacity • {intensity / duration} of exercise	Accept oxygen concentration carbon dioxide concentration Accept BMI / smoking	(2)

Question Number	Answer	Additional Guidance	Mark
2(b)(ii)	An answer that includes the following: • variable with suitable control method described (1) • results are not valid / description of expected effect on the dependent variable (1)		(2)

(Total for Question 2 = 9 marks)

